

PRD - Week 15 v4.0

Project Overview

Property Name: Hale Ōhi'a Lehua

Island: Hawai'i Island

Visitor Segment: Eco-tourists (Researchers/Visitors)

Hale Ōhi'a Lehua is located on the edges of a Ōhi'a forest on Hawai'i Island near Volcanoes National Park. The property is designed to serve two types of visitors: Research Guests who are drawn to Kīlauea for academic or professional pursuits and Leisure Guests who are attracted to Hale Ōhi'a due to its proximity to an active volcano and the unique workshops and opportunities provided by the property. I drew inspiration from the concept of Two-Eyed Seeing, which seeks to balance equally Western science and Indigenous knowledge, which I hope to reflect in the structure and presentation throughout the site.

The three pages are:

Marketing Page – the landing page that serves as an introduction to the property. Covers philosophy, and a snapshot of current conditions in Volcano National Park

Visitor Dashboard – More in-depth real-time conditions, activity suggestions, and event information

Admin Page – Interface to manage events, bookings, partner information, and adjust condition display information if needed or desired

User Personas

1. The Research Guest: These guests are coming specifically to engage with the volcano. This could be a scientist, a writer, a Hawaiian practitioner, or anyone with an academic and/or professional reason to visit Kīlauea.
2. The Leisure Guest: These guests are drawn to some aspect or offering of Hale Ōhi'a Lehua. Maybe they are intrigued by the workshops provided, or the hosted guest speaker events. It could be the amenities, proximity to forest and volcano hikes, or desire to be near others travelling for more particular reasons.
3. Property Admin: Uses the Admin Page to update event and workshop schedules, process booking and respond to inquiries, change/manage/update partner destination information, and ensure that the current volcano conditions are displaying as desired.

Marketing Page Requirements

Hero: This is to give a first impression of the page. I'd like it to have a dynamic background with media that reflects the current conditions of the volcano's activity (erupting, smoldering, dormant) along with a rotating 'Ōlelo No'eau quote.

About: Short description illustrating the property's Two-Eyed Seeing principle of equally valuing scientific and indigenous ways of thinking. Highlights proximity to crater, the curated speaker and workshop series, and resources available for both leisure and research support.

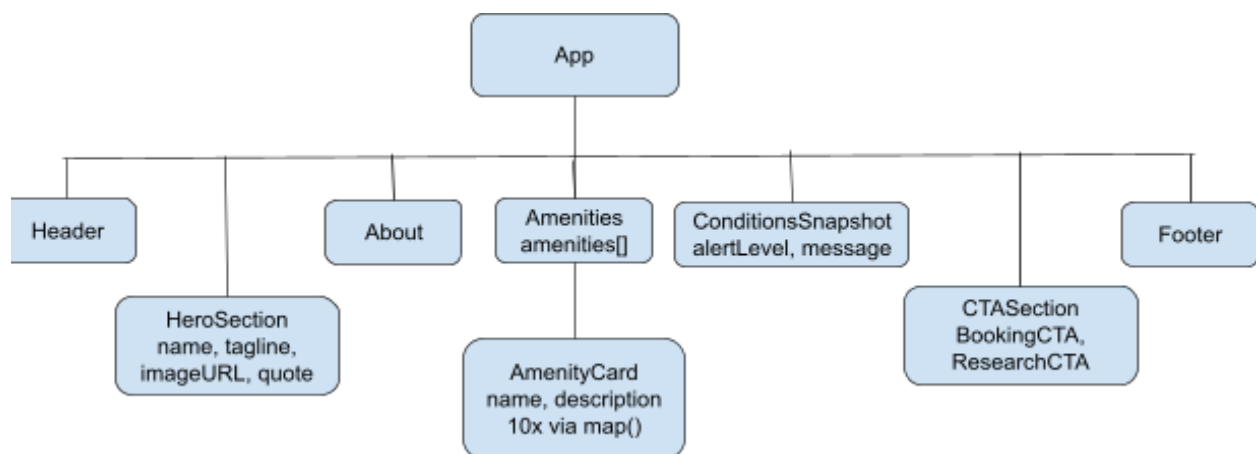
Amenities:

- Shuttle to the crater provided as well as protective gear if needed (masks, eye protection, hats, ponchos, etc)
- Workshop attendance included in accommodation price
- On-site restaurant serving farm-to-table breakfast, lunch, and dinner
- Late night lounge offering beverages and pupus – open till 10
- Clearly demarked trails through Ohi'a forest offering different ranges of time/difficulty to complete
- Complimentary laundry service
- On-site cold plunge, hottub, and sauna
- Accommodations ranging from single rooms to connected suites
- In the event of unfavorable crater conditions, shuttle services provided to select partner destinations and/or discounted car rentals available for daily/weekly rates.
- Dedicated Research Rooms available for per-hour booking

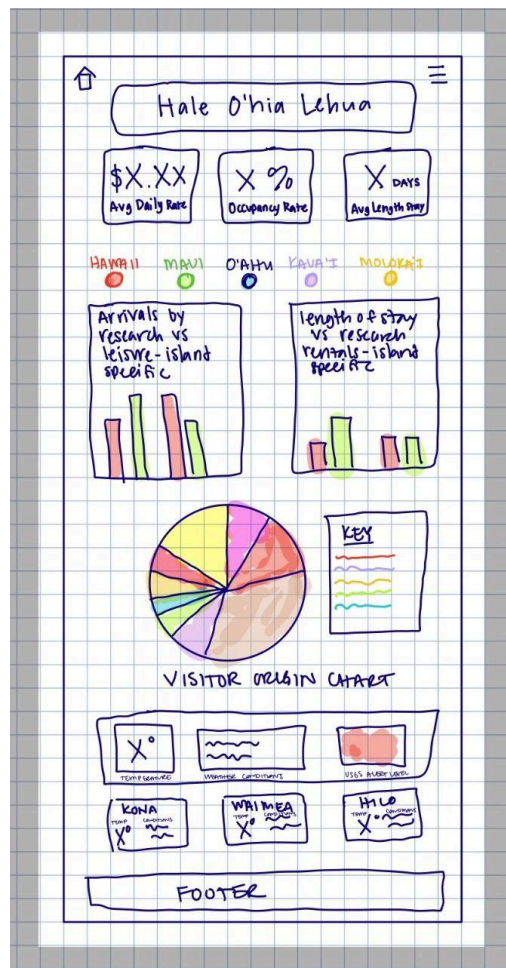
Conditions: Provide real-time information about current conditions on Kīlauea, with plain-language explanations for guests. Links through to a full Visitor Dashboard with more in-depth details.

Calls To Action: Either "Book a Stay" targeting Leisure guests, or "Plan a Research Visit" targeting Research Guests.

Component Hierarchy Diagram



Dashboard Design



IslandSelector UX

Top of page widgets (Avg Daily Rate/Occupancy Rate/Length of Stay) provide user with snapshot of statistics across the entire property.

I decided to have the filter below this be a row of toggleable buttons to allow for multiple selections for easier comparison between island statistics, using color-coding to help distinguish between different island's data. This way, if someone picks an island two bar charts automatically update: one showing how many guests arrived from this island for research vs leisure with a clear total depicted, and another bar chart showing the correlation between average length of stay and research facility rental rates (research vs leisure and research facility rental rates is synthetic placeholder data).

There's also a weather widget that depicts volcano conditions as well as current

weather conditions for Hilo, Kona, and Waimea to allow for easy comparison in case of park closures (which would result in guest redirection with partner sites). All these locations are centered specifically on the Big Island since that's the location of my property and the most pertinent for my overall property design.

The `useState` variable name is meant to store an array of selected islands rather than just one, so I will be using:

```
const [selectedIslands, setSelectedIslands] = useState([])
```

Weather Widget

For the final version of my site I envision three different versions of the weather widget, due to my property's reliance on volcanic conditions. Components will lazy-load for performance. While the API calls resolve, I'd like to display a little animated volcano eruption graphic as a placeholder that will be replaced with the conditions data once it loads.

- On the Admin page (week 13) I see a weather widget that gives a snapshot of current volcanic conditions (temperature, weather description, and USGS alert level) as well as the conditions in three other popular places to allow for easy contingency planning.
- On the Marketing page (future weeks) I see a weather widget that gives a snapshot of current volcanic conditions with a short plain-language description of what those conditions mean
- I would also like to feature a more in-depth weather page that gives a much more detailed snapshot of current volcanic conditions, a snapshot of historic volcanic activity, different warning levels for things like tephra, ash, and vog, as well as tbd information that may be of interest to research visitors.

Authentication

The method of authentication I'm using is [Passport.js](#) LocalStrategy, email plus password for login, and a single admin user.

The admin is able to view all information and property listings, see reviews and how many reviews have been received. The public cannot access any `/admin/` routes.

Protected routes: `/admin/login`, `/admin/dashboard`

Form submits to POST > /admin/login > Passport looks up email in MongoDB > bcrypt compares the password that was submitted to the stored hash > if it matches, a session is created > redirected to /admin/dashboard > if fails, return to /admin/login with error message displayed

Week15 Added feature: Google OAuth 2.0 via passport-google-oauth20 signin option for admins.

This adds flexibility - local sign-ins allow full control over credentials, however Google OAuth removes password management from the equation (which is nice for security) as well as provides a familiar, unified sign in experience for users.

User data fields: email, displayName, password (which is nullable for Google users), googleID, provider (local or google), role, createdAt

Account linking flow: check googleId first > match emails > if no email match, create new user

Session management: express-session with connect-mongo store, httpOnly cookie, saveUninitialized is set to false.

Secrets are all stored in .env that is protected by .gitignore.

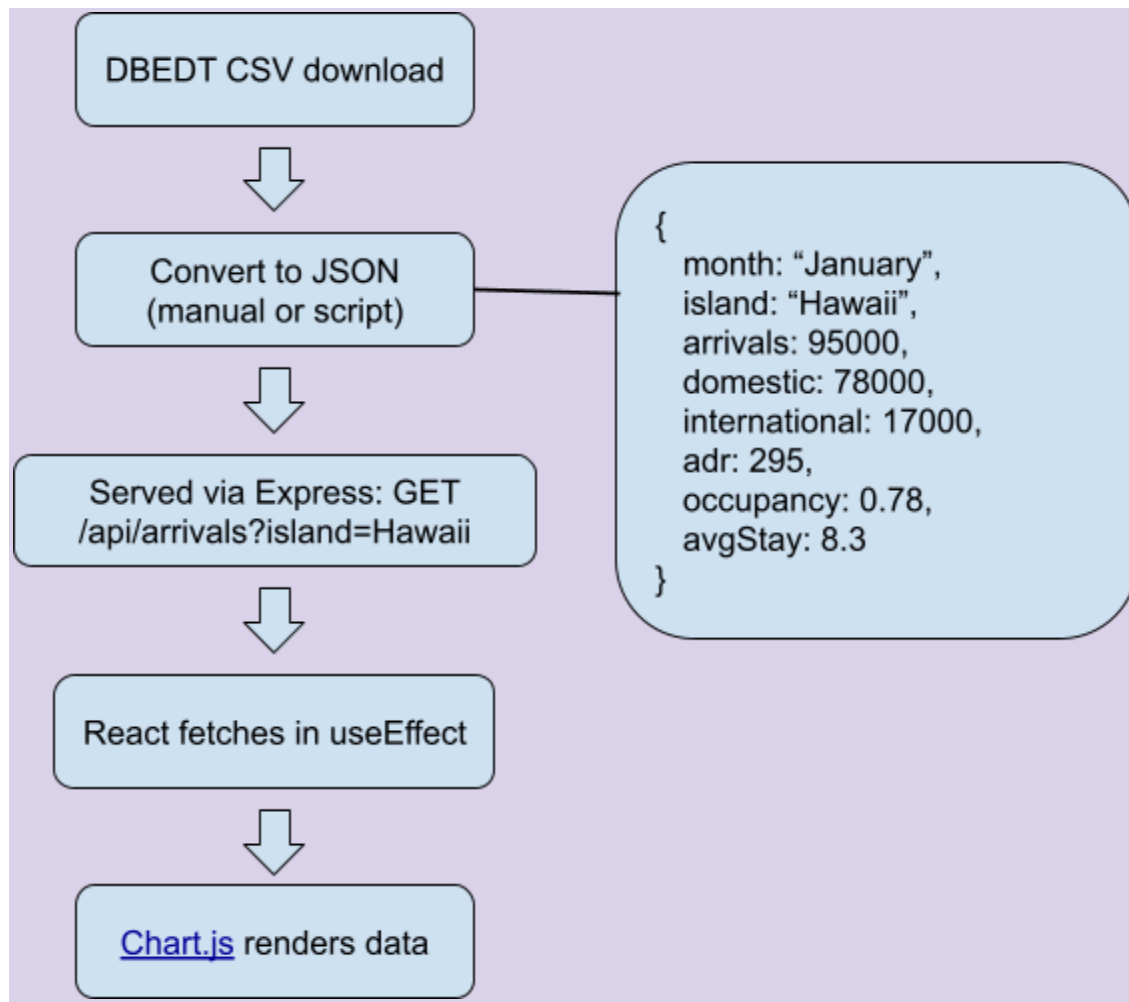
Add in Week 15: Guest Inquiry System

Add a pop-out model if someone clicks on the booking CTAs that collects information which is then fed into MongoDB and displayed on the admin dashboard.

Two forms: one for leisure guest, one for the research guest

The admin will see the results in widgets on the dashboard, incorporating the other properties I created by accident in the beginning of the project as “sister properties” the guests can also visit/stay at.

Data Architecture



Data Flow

In Week 12 the property data is hardcoded as a props object in App.jsx and passed down to child components via prop drilling.

Going forward, my plan is...:

1. MongoDB stores the property documentation (name, description, amenities)
2. Express routes via GET /properties/:id
3. React frontend uses useEffect and fetch() to request data from the Express API
4. App.jsx holds the response in state and passes it down to the child components
5. USGS HVO API supplies real-time Kīlauea alert level data to HeroSection and ConditionsSnapshot, driving the dynamic background media and plain-language condition summaries.

In Week 13 DBEDT CSV > JSON > Express /api/arrivals > [Chart.js](#)
OpenWeatherMap API > WeatherWidget component
USGS API > ConditionsSnapshot component

Going forward, my plan is.....:

1. To build out a WeatherWidget specifically designed for Marketing page
2. To build out a full conditions page that includes embedded livestream of the Volcano as well as much more detailed information on current conditions

Acceptance Criteria - Week 14 specific

1. Given user clicks on “admin” on the Marketing page the admin/login screen loads
2. Given user inputs username and password, the screen refreshes
3. Given user inputs wrong password, an error message is displayed on the login page when it refreshes
4. Given user inputs correct credentials, the admin/dashboard loads

#	Given	When	Then
1	User is on the Marketing page	They click on “admin”	The admin/login screen loads
2	User is on admin/login screen	They input their credentials	The screen refreshes
3	User is on the admin/login screen	They input the wrong password	An error message displays on the refreshed login page
4	User is on the admin/login screen	They input the correct credentials	The admin dashboard loads
5	User is on the admin/login screen	They choose to sign in with Google	The Google Authentication screen populates
6	User is on the Marketing page	They click the CTA for booking as a leisure/research guest	A modal pops up with input fields
7	User is on the Marketing page	They fill out the booking modal and hit submit	A thank you screen populates

8	User is on the Admin dashboard	They look for guest inquiries	They see submitted inquiries with tags for both leisure and research
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Week 14 - 15 Plans

Add Google OAuth to the admin login screen

Clean up any links that aren't directing where they should

Add a modal for booking that feeds information to MongoDB that is subsequently displayed on the admin dashboard.

AI Attribution

Claude was used to answer questions and get clarity about how to construct Component Hierarchy Diagram and Data Architecture Diagram